

Mend-Amar Badral

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I'm a diligent and hard-working student with a passion for facing new and exciting challenges. I believe in the power of people and collaborative spirit.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.S. in Computer Science Engineering
Budapest, Hungary
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.S. in Applied Mathematics
Ulaanbaatar, Mongolia
Thesis: [Evaluating land degradation using image processing methods](#)

EXPERIENCE

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- **Lab Leadership:** Selected as Lab Lead in final year; coordinated research progress and facilitated technical communication between 10+ students and faculty.
- **Software Engineering:** Developed and refactored image processing tools using Python, implementing segmentation methods (K-means, Moran's I) for drone imagery analysis.
- **Infrastructure:** Managed lab computing resources in a Linux environment and authored technical documentation to standardize researcher onboarding and workflows.

PROJECTS

Deep Learning Foundations Repository *Open Source*

- Reproducing key examples from Bishop's *Deep Learning* textbook to master the mathematical foundations of AI/ML from first principles.
- Published reproducible Jupyter and Pluto.jl notebook reports on GitHub, focusing on algorithmic clarity, documentation, and data visualization.

Restaurant Management System *Software Design*

- Fullstack application for the Restaurant Management System (RMS), built with Angular 20+. Backend built with NestJS, Prisma ORM, and PostgreSQL. Provides the user interface for customers, waiters, chefs, and administrators to interact with the RMS ecosystem.
- Collaborative project where I was responsible for the backend database design and management.

SKILLS & COMPETENCIES

Programming: Python, JavaScript, Julia, C/C++, R, SQL, PostgreSQL, NestJS, PyTorch
Theoretical: Applied Mathematics, Optimization, Linear Algebra, Statistics, ML
Tools: Git, Docker, Linux/Unix, L^AT_EX
Languages: English (Fluent)